

Shadow banking in India: Do bank sponsored Asset Management Companies (AMCs) perform liquidity transformation through exposure to Non-Banking Finance Companies (NBFC) in their debt oriented schemes and will this increase the systemic risk of a bank due to a possible joint exposure to NBFCs?

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Abstract:

This research attempts to address the broad issue of joint exposure that Indian banks may face through direct lending and investment exposure to Indian Non-banking finance companies (NBFC) and an indirect exposure to the same NBFC through its asset management arm (in the form of portfolio investment). This joint exposure is expected to affect the overall banking systemic risk. The first proposition is to develop a framework whether a joint exposure exists to a particular NBFC at a bank level and at the banks' mutual fund arm level (through portfolio investment). The second part is, whether the asset management arms of these banks function as shadow banks and undertake liquidity transformation by investing in NBFC floated debentures and commercial paper and how this adds to the joint exposure of the Indian banks. The third proposition of this research is to develop a framework, whether the banks stress testing levels may be altered significantly in case a joint exposure exists. In this study, we have developed the second proposition: do mutual funds undertake liquidity transformation and in the process have created a framework and have tested whether liquidity transformation occurs. Through a pilot study using the overall debt schemes we find some evidence that suggests liquidity transformation is performed. We intend to use data from Reserve Bank of India (RBI) website, Securities Exchange Board of India (SEBI), Association of Mutual Funds India (AMFI), Micro Finance Institutions Network (MFIN), monthly portfolio disclosure of asset management companies' schemes and annual and quarterly reports of NBFCs and banks during the period 2012 to 2017.

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Introduction:

This study begins with describing how the global financial markets suffered due to the sub-prime crisis in 2007-08 as a number of financial institutions (such as mutual funds) faced an exposure to debt instruments that were backed by poor quality mortgage assets. These financial institutions had undertaken maturity and liquidity transformation similar to a traditional bank. These institutions were suggested to have functioned like a “shadow bank”. These shadow banking activities went undetected under the regulatory framework that generated a fall out. This created a very high systemic risk for the whole financial system. Although shadow banking is highly pervasive in the developed countries such as U.S. and Europe, but also it is extremely prevalent in the emerging economies since these economies have a large population that has no access to traditional banking. An emerging economy like India has Non-banking Finance Companies (NBFCs) that function as shadow banks and lend to corporate and retail borrowers with reasonable credit history function (Acharya et.al, 2013).

These NBFCs act as a complement to the traditional banking system. However, they aren’t regulated as closely as Indian banks. These NBFCs loan book growth has been outperforming the Indian banks loan book growth over the past few years. This NBFC loan book growth is being funded through bank loans and asset management companies (AMC). However, lately NBFCs reliance on bank funding has reduced whereas AMC funding (through investment in NBFC floated commercial papers and debentures) has increased as banks are marred with non-performing assets (NPA). More than 17-44% of the Net Asset Value (NAV) of these debt-oriented schemes of AMCs is in NBFC floated securities as. Owing to this shift in the funding exposure of AMCs to NBFC floated instruments a question that arises is: *Are these AMCs functioning as credit intermediaries to NBFCs? If yes, then does this serve as a concern?*

Majority of these AMCs are bank sponsored. Hence, this may increase the level of systemic risk on three fronts. A) bank level B) AMC level and C) NBFC level. Since, banks and majority of these NBFCs already undertake credit intermediation in the financial system. This AMCs funding to NBFCs may also undertake some form of credit intermediation.

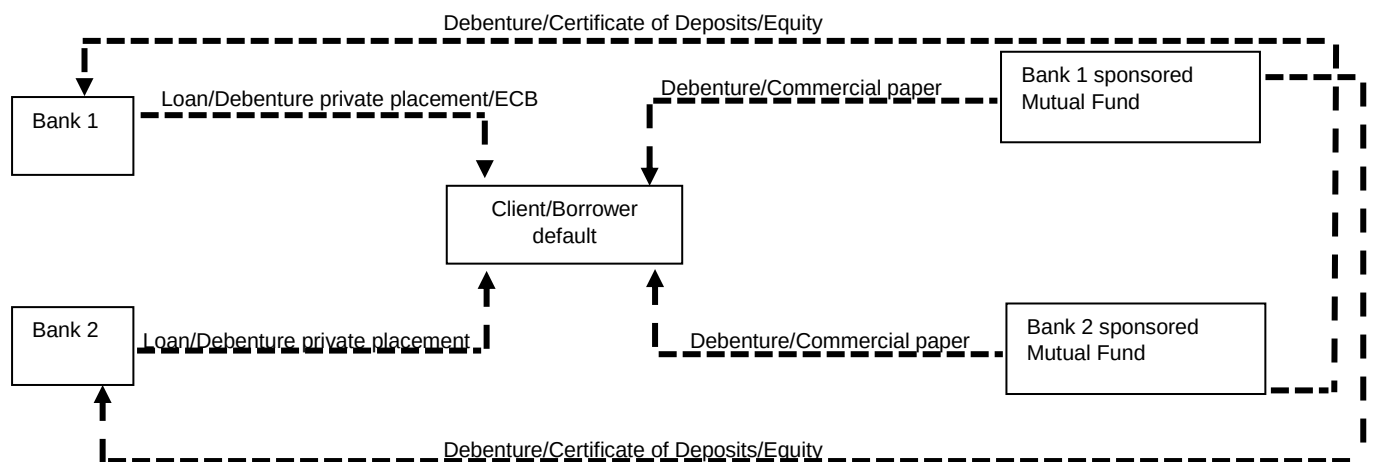
If this credit intermediation occurs this makes the banks take a dual exposure to NBFCs through 1) Direct lending and investment 2) Investment through AMC held portfolio.

The Reserve bank of India (RBI) duly monitors the NBFC exposure limit at the bank and the AMC level separately. To some extent the Basel III pillar of disclosures deliberates on the consolidated risk for the bank and the bank sponsored AMC. However, these AMCs may be indulging in liquidity transformation on behalf of their parent banks as they fund NBFC floated securities using investor money. Second, these AMCs may have an exposure to other banks – who may in turn have an exposure to these NBFCs not only at the bank level but also at their AMC level. This may create an added risk for a bank if the NBFC faces stress. Further in this introduction, we continue to elaborate four scenarios in a diagram that explains how this NBFC exposure increases risk in a financial system. The first two scenarios contain the banking and the AMC level framework. However, the latter two scenarios introduce the NBFC exposure. In scenario 1, we begin with a simple example: Jindal Steel and Power Limited (JSPL - Borrower) has an overall debt estimated at INR.458.8 billion³. State Bank of

³ Consolidated Balance Sheet, Financial Year 2016-2017, Page 20, or Economic Times, Lenders to take JSPL account off bad loan list, 20 Jan 2018, available on the internet at,

India (SBI) exposure is at INR.80 billion. In 2016, JSPL debt worth INR.71.25 billion⁴ was to undergo restructuring. In this, SBI had an exposure of INR.30 billion followed by Axis Bank, ICICI Bank and IDBI Bank. ICICI Bank has an exposure of INR.23.54 billion, in loans and external commercial borrowings (ECB), which was INR.25.03 in March 2016. JSPL also has INR.3 billion worth debentures that were privately placed with ICICI Bank. As on 31 March 2017, JSPL owed 91.53 million of the 5.55 billion (under defaults) to ICICI bank. On the mutual fund front, ICICI Prudential had an exposure of INR.5 billion⁵ to JSPL debt under its mutual fund schemes with INR.1.52 billion exposure to its scheme ICICI Prudential Regular Savings Fund (2.91% of the INR.52.45 billion AuM on 31 March 2016). Overall, ICICI has an exposure to JSPL is INR. 32.45 billion on March 2017. Under the Basel III banks provide for the exposure to JSPL through ICICI bank and under its mutual fund arm – ICICI Prudential Mutual Fund. However, when allied banks such as Axis, SBI, HDFC and IDBI also have an exposure to JSPL. Now ICICI mutual fund arm may actively subscribe to debt instruments floated through these allied banks or to equity of these allied banks. So, does the Basel III norms include this aspect, what impact will the whole system have on the ICICI bank and also on the other banks because of this joint exposure to JSPL. This gives rise to Scenario 1.

Scenario 1: Bank 1 exposure to a) Borrower default, b) Bank 1 sponsored Mutual Fund exposure to borrower default and c) other banks floated debt or equity



- Bank 1 has exposure to Borrower through loans/debtenture private placement
- Bank 1 sponsored Mutual fund has exposure to debtenture/commercial paper issued by borrower
- Bank 2 has exposure to Borrower through loans/debtenture private placement
- Bank 2 sponsored mutual fund has exposure to debtenture/commercial paper issued by borrower

<https://economictimes.indiatimes.com/markets/stocks/news/lenders-to-take-jspl-account-off-bad-loan-list/articleshow/62578088.cms>, accessed 14 March 2018.

⁴ JSPL gets banks' nod to recast INR.7125 crore debt under 5/25 scheme, available on the internet at, <https://www.moneycontrol.com/news/business/companies/jspl-gets-banks-nod-to-recast-rs-7125-cr-debt-under-525-scheme-937778.html>, accessed on 14 March 2018

⁵ Business Standard, JSPL downgrade puts Franklin and ICICI Prudential in a spot, 18 Feb 2016, available on the internet at, http://www.business-standard.com/article/markets/jspl-downgrade-puts-franklin-and-icici-prudential-in-a-spot-116021700900_1.html, accessed on 14 Mar 2018

- Bank 1 sponsored mutual fund arm has exposure to Bank 2 issued debentures/ Certificate of Deposits/Equity

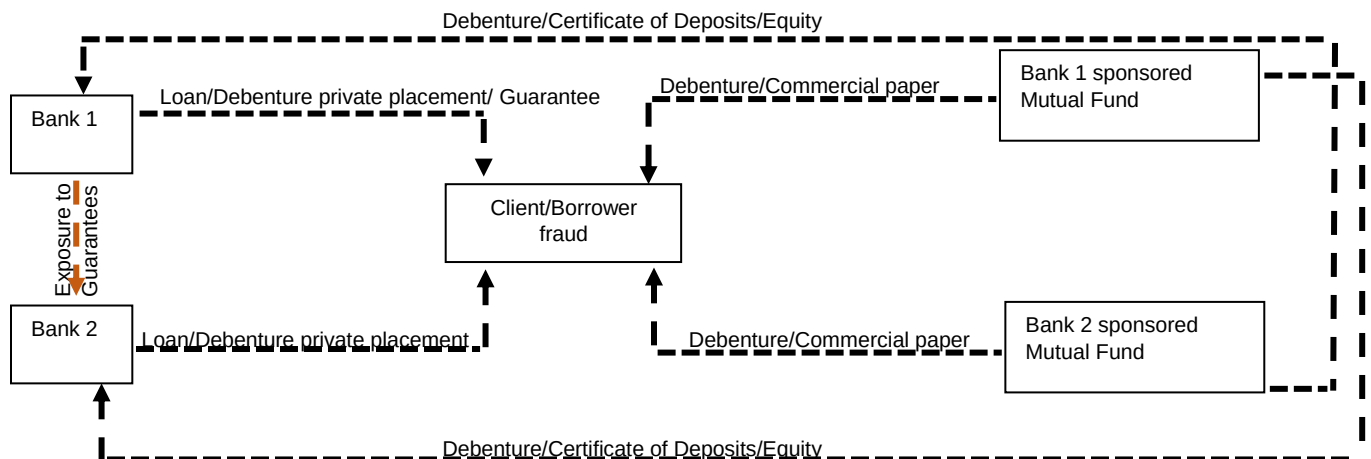
Total exposure of Bank 1 to the borrower default = Direct lending exposure to borrower + Bank 1 sponsored Mutual fund exposure + Indirect Exposure through investment in Bank 2 + Investment in Bank 2 sponsored mutual fund + Indirect exposure of Bank 1 sponsored mutual fund arm to Bank 2.

If a borrower fails to repay; and if bank 2 has substantial exposure to borrower. The bank 1 is not only affected through direct lending exposure and its mutual fund arm exposure to the borrower but also through its exposure to bank 2 debt and equity and to bank 2 sponsored mutual fund arm.

Now, we consider Scenario 2 in which bank guarantees are also included.

Scenario 2: Bank 1 exposure to a) Borrower default, b) Bank 1 sponsored Mutual Fund exposure to borrower default, c) other banks floated debt or equity, d) bank guarantee issued on behalf of borrower

The recent Nirav Modi fraud through Letter of Undertaking (LoU) worth INR.126 billion issued by Punjab National Bank (PNB) helped us design this scenario 2, where Bank 1 also has an exposure to bank guarantees issued to borrowers.

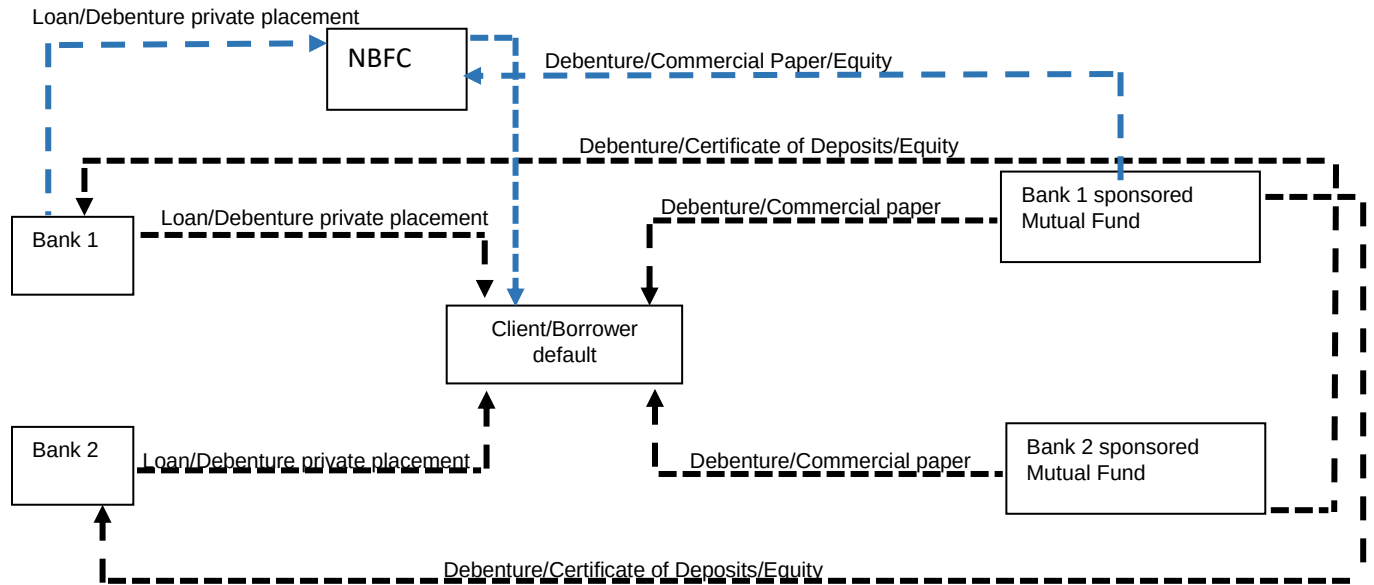


- Bank 1 has exposure to Borrower through loans/debenture private placement
- Bank 1 has exposure to Borrower guarantee to Bank 2
- Bank 1 sponsored Mutual fund has exposure to debenture/commercial paper issued by borrower
- Bank 2 has exposure to Borrower through loans/debenture private placement
- Bank 2 sponsored mutual fund has exposure to debenture/commercial paper issued by borrower
- Bank 1 sponsored mutual fund arm has exposure to Bank 2 issued debentures/ Certificate of Deposits/Equity

Total exposure of Bank 1 to the borrower default = Direct lending exposure to borrower + Bank 1 sponsored Mutual fund exposure + Indirect Exposure through investment in Bank 2 + Investment in Bank 2 sponsored mutual fund + Indirect exposure of Bank 1 sponsored mutual fund arm to Bank 2 + Bank 1 exposure to borrower guarantee to Bank 2.

The third scenario introduces NBFCs exposure to the same client. Here, Bank 1 is exposed not only to the NBFC because banks serve as direct lender to NBFCs but also through its mutual fund arm.

Scenario 3: Non-banking Finance exposure, Bank 1 exposure to a) Borrower default, b) Bank 1 sponsored Mutual Fund exposure to borrower default, c) other banks floated debt or equity, d) NBFC and e) Bank 1 mutual fund exposure to NBFC floated debt or equity

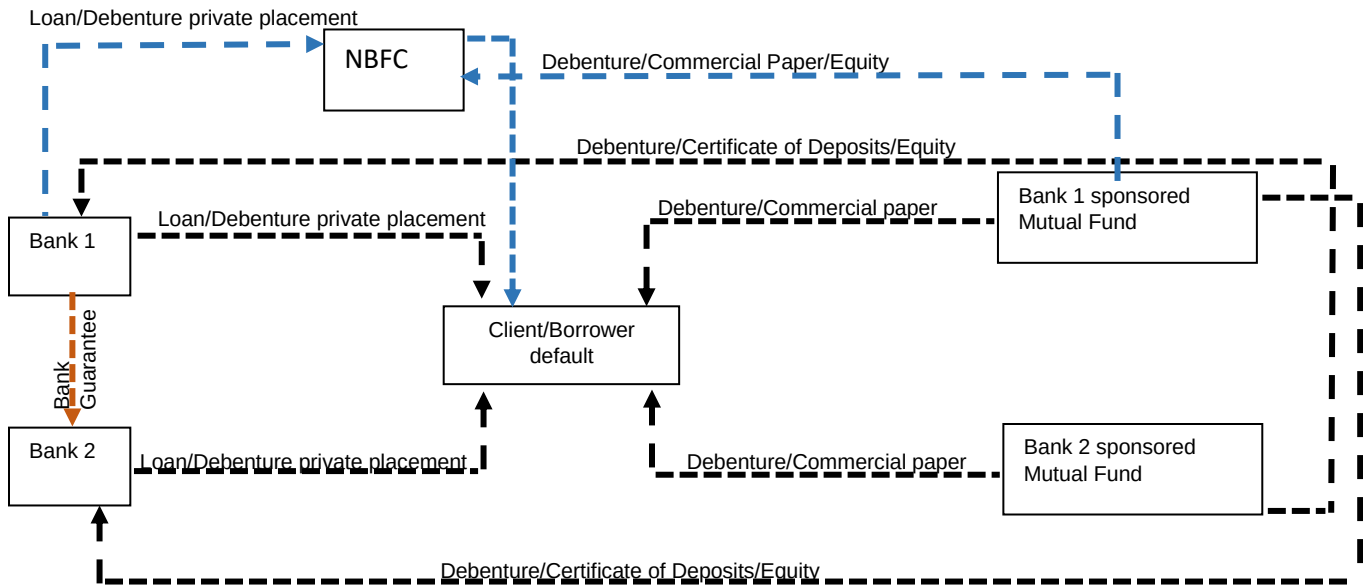


- Bank 1 has exposure to Borrower through loans/debenture private placement
- Bank 1 sponsored Mutual fund has exposure to debenture/commercial paper issued by borrower
- Bank 2 has exposure to Borrower through loans/debenture private placement
- Bank 2 sponsored mutual fund has exposure to debenture/commercial paper issued by borrower
- Bank 1 sponsored mutual fund arm has exposure to Bank 2 issued debentures/ Certificate of Deposits/Equity
- Bank 1 has exposure to NBFC that undertakes direct lending to borrower
- Bank 1 sponsored mutual fund has exposure to NBFC that invests in borrower debentures/commercial paper/equity

Total exposure of Bank 1 to the borrower default = Direct lending exposure to borrower + Bank 1 sponsored Mutual fund exposure + Indirect Exposure through investment in Bank 2 + Investment in Bank 2 sponsored mutual fund + Indirect exposure of Bank 1 sponsored mutual fund arm to Bank 2 + Bank 1 direct lending exposure to NBFC which lends to borrower + Bank 1 sponsored mutual fund has exposure to borrower debentures/commercial paper/equity.

The fourth scenario builds up from the third scenario and adds an element of bank guarantee

Scenario 4: Non-banking Finance exposure, Bank 1 exposure to a) Borrower default, b) Bank 1 sponsored Mutual Fund exposure to borrower default, c) other banks floated debt or equity, d) NBFC, e) Bank 1 mutual fund exposure to NBFC floated debt or equity and f) bank guarantee



- Bank 1 has exposure to Borrower through loans/debenture private placement
- Bank 1 sponsored Mutual fund has exposure to debenture/commercial paper issued by borrower
- Bank 1 exposure to bank guarantees issued on behalf of borrower
- Bank 2 has exposure to Borrower through loans/debenture private placement
- Bank 2 sponsored mutual fund has exposure to debenture/commercial paper issued by borrower
- Bank 1 sponsored mutual fund arm has exposure to Bank 2 issued debentures/ Certificate of Deposits/Equity
- Bank 1 has exposure to NBFC that undertakes direct lending to borrower
- Bank 1 sponsored mutual fund has exposure to NBFC that invests in borrower debentures/commercial paper/equity

Total exposure of Bank 1 to the borrower default = Direct lending exposure to borrower + Bank 1 sponsored Mutual fund exposure + Bank guarantee issued to Bank 2 on behalf of borrower + Indirect Exposure through investment in Bank 2 + Investment in Bank 2 sponsored mutual fund + Indirect exposure of Bank 1 sponsored mutual fund arm to Bank 2 + Bank 1 direct lending exposure to NBFC which lends to borrower + Bank 1 sponsored mutual fund has exposure to borrower debentures/commercial paper/equity.

The scenario 3 and 4 leads us to three unanswered questions:

- a) Does joint/dual exposure to NBFCs exist in the present Indian financial system?
- b) Do bank run AMCs through their open-ended debt-oriented schemes indulge in liquidity transformation?
- c) Do banks face exposure to other banks and their AMCs in the system that may also have an exposure to these NBFCs? Do we account for the risk under this exposure?
- d) Does the banks systemic risk increase due to the joint/dual exposure to NBFCs?

Previous studies have focused on liquidity transformation in AMCs (Sunderam and Cherenko, 2016, redemptions and fund manager cash holdings (Shek, Shim and Shin, 2017; Morris, Shim and Shin, 2017; Jiang et al., 2017), systemic risk in the financial system (Acharya et al., 2017). Acharya et al. (2013) explains through his study how NBFCs function as complements to the existing banking system in India. However, these studies have not considered how dual exposure to a credit intermediation firm such as a NBFC at the bank level and through its mutual fund arm may result into higher systemic risk for the entire banking system. Another part is they have not reflected upon the interdependency between the NBFCs and banks that may have exposure to a common borrower and how it also adds to the systemic risk.

This study is categorized into five sections. The first section covers the overview of shadow banking in the global and Indian market, the second section covers the overview of non-banking finance companies (NBFC) in India, the third section covers the NBFC loan growth, the fourth section covers how the NBFCs have been funding their loan growth and the fifth section describes the financial interdependency among banks and asset management companies, the section six gives a brief about the data sources, the possible methodology that will be employed and the final section provides a brief about the pilot study conducted.

1. Shadow Banking: Overview

Shadow banking has been an area of interest post the 2007-2008 US financial crisis. In the developed nations such as US and Europe, money market mutual funds (MMMF) and debt mutual fund schemes were considered as shadow banks who undertook the function of credit intermediation without having access to central bank liquidity facilities and deposit insurance guarantee facilities (See: Chernenko and Sunderam, 2014; Duygan-Bump et.al., 2013;). These mutual fund schemes also fell prey to the sub-prime mortgage default crisis that occurred in the US during 2007-08. A number of these mutual fund schemes had an exposure to debt instruments that were backed by poor quality assets such as residential and commercial mortgages. When the crisis occurred the mutual fund schemes were faced with heavy investor redemption demands and were unable to liquidate the underlying sub-prime mortgage assets backed by securities (commercial paper, asset backed securities, bonds). The redemptions subjected mutual fund schemes to a fire sale or suspension of investor redemptions till a suitable action plan was ascertained.

An example of a MMMF is the Reserve Primary Fund that held US\$785 million in Lehmann issued securities and became illiquid when the fund was unable to meet investor redemption demand⁶.

⁶ Securities Exchange Commission website, Reserve Primary Fund Distributes Assets to Investors, 29 Jan 2010, available on the internet at, <https://www.sec.gov/news/press/2010/2010-16.htm>, accessed on 20 Jan 2018

Three European debt mutual fund schemes⁷ such as BNP Paribas ABS Eonia⁸, BNP Paribas Euribor (French domiciled) and Parvest Dynamic ABS (Luxembourg domiciled) had an exposure of 35% of their assets to US sub-prime Asset Backed Securities (ABS) in July 2007 suspended redemptions for 20 days from 8th Aug 2007 to 28th Aug 2007. The concept of shadow banking has gradually evolved where, credit intermediation takes place outside the conventional banking system. In this system, a non-banking financial institution engages in liquidity, credit and maturity transformation. However, unlike a bank these non-banking financial institutions don't have access to the central bank liquidity facilities and deposit insurance guarantee facilities like a bank.

These U.S and European based non-banking financial institutions (such as money market mutual funds) faced systemic risks which surfaced during the 2008 financial crisis and went undetected under the regulatory framework. To resolve this issue the US and European financial markets introduced a number of reforms in the industry such as increasing disclosures of monthly portfolio holdings, introducing floating net asset value (NAV) for institutional floated funds, enhancing diversification and setting single institution exposure limit to 10%-15%, imposing a 2% redemption fees on liquidation in case the weekly liquid assets dips below 30% of its total assets etc⁹. Similarly, the Financial Stability board (FSB) has actively undertaken efforts to monitor and measure shadow banking so that the unforeseen systemic risks are minimized and opportunities created through shadow banking benefit the overall financial system.

Shadow banking is more prominent in the US system and less pervasive in the European region. However, in emerging economies such as India; non-banking financial institutions (shadow banks) serve as complements to the existing banking institutions due to skewed banking penetration levels across the country. There exists a weak positive correlation between the banking and non-banking financial institutions (NBFIs) loan growth during the period March 2014 to Sep 2017 indicating that these NBFIs serve as complements to the existing banking system (See Diagram 1). In India, 58.7% of the Indian households (India has 246.74 million households: Census data 2011¹⁰) avail banking facilities. Banking penetration is higher in the urban areas (67.76% of Indian households) whereas rural areas (54.4% of Indian households) have limited banking penetration. Global Findex data¹¹ suggests that banking penetration is at 53% in 2014 i.e. 468.81 million account holders (of which 175 million people opened accounts during the year 2011 to 2014). However, 43% of these accounts remain dormant¹². To fill the gap of these banking services NBFIs support market liquidity, maturity transformation and risk sharing (Ghosh et al., 2012). In India these NBFIs that fill the banking gap are in the form of Non-Banking Financial Companies (NBFC) and collective investment vehicles such as money market funds, fixed income funds real estate

⁷ Asset Back Securities (ABS) Fund by definition must invest at least 80% of their assets in securitized assets such as asset backed securities.

⁸ BNP Paribas Documents, Background Information on suspension and reopening of ABS funds in August, available on the internet at, <https://invest.bnpparibas.com/sites/default/files/documents/5761.pdf>, accessed on 20 Jan 2018.

⁹ SEC adopts Money market Fund reform rules, 2014, available on the internet at, <https://www.sec.gov/news/press-release/2014-143>, accessed on 20 Jan 2018

¹⁰ Census India 2011, Number of Households availing banking services and number of households having each of the specified assets, available on the internet at, <http://www.censusindia.gov.in/2011census/Hlo-series/HH12.html>, accessed on 20 Jan 2018

¹¹ World Bank Global Financial Inclusion Index 2014, South Asia, Key Indicators, available on the internet at, <http://datatopics.worldbank.org/financialinclusion/country/india>, accessed on 20 Jan 2018

¹² The Indian Express, 10 Nov 2015, In India bank account penetration surges, but 43% dormant, available on the internet at, <http://indianexpress.com/article/india/india-news-india/in-india-bank-account-penetration-surges-but-43-dormant/>, accessed on 20 Jan 2018

funds and securitization- based credit intermediation like securitization vehicles and structured finance vehicles. Acharya et. al. (2013) through their study suggested that these NBFCs in India operate as shadow banks and undertake liquidity, maturity transformation and leverage activities similar to banks. Shadow banking in India is estimated at USD436 billion in 2015 and forms 14% of the total financial assets¹³. Among Brazil, Russia, India, China and South Africa (BRICS) nations, India forms the third largest shadow banking sector.

¹³ FSB, Global Shadow Banking Monitoring Report, 2016

2. Overview of Non-Banking Finance Companies (NBFC) in India

NBFCs are registered under The Companies Act, 1956 and engage in the business of loans and advances, leasing, hire-purchase, insurance business, chit funds, trading, acquisition of securities such as shares, bonds, debentures and others¹⁴. In March 2017, there are 11,522 NBFCs of which 220 are NBFC Non-deposit taking Systematically Important (NBFC ND SI). These 220 NBFC ND SI form 86% of the total assets of the NBFC sector¹⁵. Table 1, displays the asset size and the number of NBFC ND-SI from 2014 till 2017. We observe in Table 1 that the number of NBFC ND SI have more than halved from 493 in 2014 to 220 in 2017. This reduction was attributed to a revision in the regulatory guideline that increased the threshold asset size requirement of these NBFC ND SI from INR 1 billion to INR 5 billion (See Table No. 1) thereby disqualifying a number of the NBFCs with the prior granted status of NBFC ND SI.

Public limited companies' form 47.7% of the total NBFC ND SI whereas private limited companies contribute for the remainder 45.45%. The asset size for private limited companies reduced from INR.6,856 billion to INR.2,369 billion and the asset size increased for the public listed companies from INR.1,705 to INR.8,268 billion. The reason behind an increase in the asset base of public listed firms can be attributed to the initial public offerings (IPO)¹⁶ launched by MAS Financial Services, Ujjivan Financial Services and a few others and the capital raised through debentures and commercial paper borrowings.

Table No. 1: Number and Asset size of NBFC ND-SI

Ownership	March 2014		March 2015		March 2016		March 2017	
	No.	Asset Size (In INR bn)	No.	Asset Size (In INR bn)	No.	Asset Size (In INR bn)	No.	Asset Size (In INR bn)
Total	493	12742	471	15232	220	14833	220	16917
Government	15	4181	15	5337	15	5765	15	6280
Non-government	478	8561	456	9895	205	9068	205	10637
Public Ltd. (1)	252	1705	243	2120	105	2026	105	8268
Private Ltd. (2)	226	6856	213	7775	100	7041	100	2369

Source: Created by authors, Reserve Bank of India, Financial Stability Report, Report on Trend and Progress of Banking in India 2016-17, Pg no. 142

¹⁴ Reserve Bank of India website, Frequently Asked Questions, Definition of NBFC, 10 Jan 2017, available on the internet at, <https://www.rbi.org.in/Scripts/FAQView.aspx?Id=92>, accessed on 26 Jan 2018.

¹⁵ FSB 2017 report, available on the internet at, <https://rbidocs.rbi.org.in/rdocs/Publications/PDFs/7CHA7201709E69A4012B24EE590B260B1F151BB97.PDF>, accessed on 15 Jan 2017.

¹⁶ MoneyControl, NBFCs laugh their way to the bank as rich investors try to cash in on IPOs, available on the internet at, <http://www.moneycontrol.com/news/business/ipo/nbfc-laugh-their-way-to-the-bank-as-rich-investors-try-to-cash-in-on-ipo-mania-2328453.html>, accessed on 15 Jan 2017.

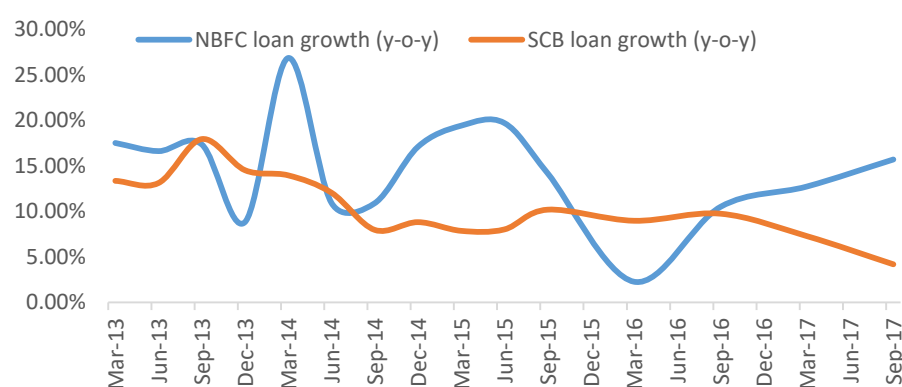
3. NBFCs loan growth higher than Scheduled Commercial Banks

The Indian NBFC sector has witnessed loan growth from INR.7,233.52 billion in March 2012 to INR.14,846.00 billion in March 2017, a 105.24% growth over the past five year period. This complementary financing system has been outperforming the loan growth of scheduled commercial banks (SCB)¹⁷. These SCBs cater to 85% of the banking population. However, over the past few years since 2013 onwards the year on year loan growth of the NBFC sector has outperformed that of the SCBs.

Diagram No. 2 displays the loan outstanding growth rates for both NBFC sector and Scheduled Commercial Banks (SCBs) for the period March 2012 to September 2017 on a quarterly basis. SCB loan growth on a quarter to quarter basis has been on the lower side of 1-5%, whereas NBFC loans and advances portfolio growth has moved in the range of -2% to 14%. Eleven of the eighteen quarters over the period March 2013 to September 2017, NBFC loan outstanding growth rate was higher than the SCB loan outstanding growth rate. For 11 of the 15 quarters the NBFC loan y-o-y growth has outperformed the SCB loan growth.

A few reasons attributable to the growth in the NBFC loan outstanding are: 1) The inability of SCBs to undertake higher lending activities due to the huge pile of non-performing assets (NPA) that have been accumulated since the financial crisis in 2007-08 in the banking system. 2) The growth in NBFC loan may be attributed to the revised norms, where Reserve Bank of India increased the borrowing limit for individual borrowers from INR.50,000 to INR.100,000 in April 2015¹⁸. Hence, these two reasons have mostly contributed the NBFCs growth as they have partially substituted for the SCBs; since the SCB credit offtake has reached an all-time low.

Diagram No. 2: NBFC loan growth Vis a Vis SCB loan growth (y-o-y)



¹⁷ RBI website, SCB definition: All banks included in the second schedule to the Reserve Bank of India Act, 1934. They are categorized into five different categories: 1) State Bank and its associates, 2) Nationalised banks, 3) Private sector banks, 4) Foreign banks and 5) Regional Rural Banks.; available on the internet at, <https://www.rbi.org.in/scripts/PublicationsView.aspx?id=14655>, accessed on 28 Jan 2018.

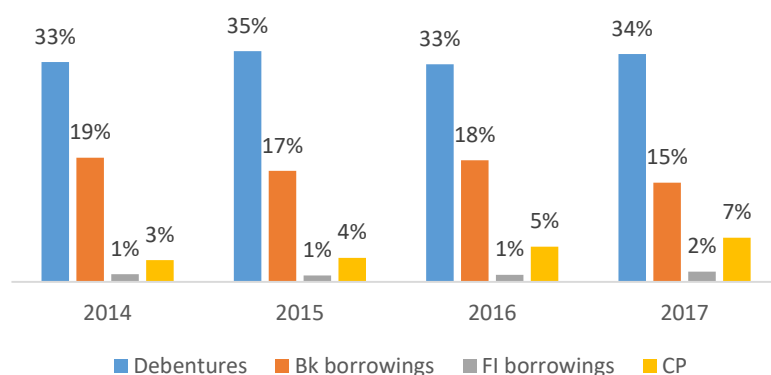
¹⁸CARE Ratings, NBFC-Microfinance Institutions- Update on Regulatory Changes, April 2015, available on the internet at, <http://www.careratings.com/upload/NewsFiles/SplAnalysis/Microfinance%20Sector%20-%20Update%20on%20Regulatory%20Changes%20docx.pdf>, accessed on 22 Jan 2018

Source: Created by the authors, Reserve Bank of India, Financial Stability report

4. How are these NBFCs funding their loan growth?

A NBFC funds its lending activities through 70% borrowing on its balance sheet. The funding composition of a NBFC typically forms debentures, bank borrowing, commercial paper, borrowing from other financial institutions etc. Diagram No.2 presents the funding composition of Indian NBFCs over the period 2014 to 2017. Debentures and banks (49% of total NBFC total assets) have been a major source of funding for NBFCs. A decline is witnessed in the bank borrowing as a percentage of total NBFC assets from 22% to 16% of the NBFC balance sheet (INR.19,671 billion) in March 2017. This is in light of the slowing banking system that is laden with non-performing assets as a result of which banks are unable to make new loans. This decline in bank borrowing is being offset by an increase in the NBFC floated commercial papers and debentures (See: Diagram No.2)¹⁹.

Diagram No.2: NBFC funding composition as a percentage of total NBFC assets



Source: Created by authors, Reserve Bank of India, Financial Stability Report, Report on Trend and Progress of Banking in India 2016-17, Pg no. 144

NBFC floated commercial paper outstanding as a percentage of total commercial paper outstanding was around ~32% in March 2017²⁰. NBFC floated commercial papers formed roughly 49% of the total commercial papers outstanding in March 2014. During this year, NBFCs with a high credit rating could raise funds through CPs at a rate that was 150 basis points lower than the average bank base rates²¹. Again around 2016-2017, the CP share in the total CP outstanding increased from 24% to a steady 32% owing to increased demand from AMCs and a funding rate

¹⁹ The Economic Times, NBFCs lower exposure to banks rely on commercial papers and bonds, available on the internet at, <https://economictimes.indiatimes.com/industry/banking/finance/nbfc-lower-exposure-to-bank-loans-rely-on-commercial-papers-and-bonds/articleshow/45697536.cms>, accessed on 23 Jan 2018

²⁰ Zee Business, 22 Aug 2016, Why are companies flocking to 'unsecured' commercial papers?, available on the internet at, <http://www.zeebiz.com/india/news-why-are-companies-flocking-to-unsecured-commercial-papers-5238>, accessed on 24 Jan 2018.

²¹ The Economic Times, NBFCs lower exposure to bank loans, rely on commercial papers and bonds, 31 Dec 2014, available on the internet at, <https://economictimes.indiatimes.com/industry/banking/finance/nbfc-lower-exposure-to-bank-loans-rely-on-commercial-papers-and-bonds/articleshow/45697536.cms>, accessed on 26 Jan 2018.

lower than the average bank base rates²². Hence, CP funding has witnessed a phenomenal increase as a portion of NBFC funding during the period 2012 to 2017.

These NBFC floated commercial papers have investor participation from asset management companies (AMCs), banks and insurance companies. AMCs are major participants of these commercial papers²³. The net exposure²⁴ that AMCs have to NBFCs has steadily increased year on year from INR. 83 billion in March 2012 to INR.1489 billion in March 2016²⁵.

Indian banks direct exposure to the NBFC sector is INR.4.9 trillion²⁶ of which NBFC floated commercial paper and debentures forms 40% (of the total INR.4,996 billion) as on March 2017. Bank loans in the form of term loans and working capital loans still form the bulk of the exposure (55% of INR.4996 billion).

Banks may face a twin exposure to a particular NBFC through:

- 1) Direct exposure: Loans and investments on their bank balance sheet
- 2) Indirect exposure: Portfolio investment in equity, debentures and commercial paper of a NBFC through the bank sponsored mutual fund arm.

However, the indirect exposure through the bank run AMC to the same NBFC is not fully accounted for in the banking system. RBI sets limits on the exposure taken by banks²⁷ and AMCs²⁸ independently to NBFCs. However, the RBI doesn't provide any guidelines that cover joint exposure to a particular NBFC undertaken by a bank and the same bank run AMC.

Although Basel III disclosure considers the bank risk at the consolidated level for both the parent bank and the bank-run AMC. However, the business norms for NBFC lending are divergent from banks, where a bank reports a non-performing asset (NPA) after 90 days and a NBFC reports it after 180 days. Hence, a bank may register the NBFC loan as NPA earlier than a NBFC may report it. In such a scenario, the bank may provide for the NBFC loan default, however at the AMC level that subscribes to the NBFC commercial paper and debentures. There may be limited reflection

²² India Infoline, Issuance of commercial paper rises, 03 Mar 2017, available on the internet at, https://www.indiaonline.com/article/news-top-story/issuance-of-commercial-paper-rises-117030300249_1.html, accessed on 26 Jan 2018.

²³ The Financial Express, Yields on NBFC commercial papers rise ahead of IPOs, 8 Sep 2017, available on the internet at, <http://www.financialexpress.com/market/yields-on-nbfc-commercial-papers-rise-ahead-of-ipos/846066/>, accessed on 26 Jan 2018

²⁴ The AMCs included here are Bank run, Financial Institutions run and Corporate run AMCs

²⁵ Reserve Bank of India, Financial Stability Report, 2013, 2014, 2015, available on the internet at, <https://rbi.org.in/Scripts/FsReports.aspx>, accessed on 23 Jan 2018

²⁶ Bank exposure includes scheduled commercial banks such as private sector banks and public sector banks and foreign banks.

²⁷ Banks have a direct exposure limit for the NBFC lending activities and investment activities capped at 10% to 15% (of the banks' capital funds as per the last audited financial statements) for a single NBFC/NBFC Asset Financing Company/ NBFC Infrastructure Finance Company. RBI exposure limit to a single NBFC, available on the internet at, https://www.rbi.org.in/scripts/BS_ViewMasCirculardetails.aspx?id=9875, accessed on 19 Jan 2018.

²⁸ AMCs have an exposure limit of 25-30% to the NBFC sector and a single issuer limit of 10-15% of the net asset value (NAV). Live Mint, SEBI relaxes debt fund exposure limit for housing finance companies, 11 Aug 2016, available on the internet at, <http://www.livemint.com/Money/kJdEWM3z3hecPWSwKMvZtK/Sebi-relaxes-debt-fund-exposure-limit-for-housing-finance-co.html>, accessed on 23 Jan 2018.

on their asset quality. In this setup, the banks overall systemic risk would tend to increase manifold. The direct and indirect exposure to a particular NBFC may be well within the RBI prescribed individual limits for a bank and AMC. However, the risk generated through this total exposure (direct and indirect) may be well beyond the banks' capacity to sustain the impact of default on the part of an NBFC. Authorities may wish to consider both direct and indirect exposure to a particular NBFC and what impact it might probably have on the parent bank using various stress testing scenarios. With this background we arrive at our first proposition for this research:

Proposition 1: Does joint exposure exist for a bank and its bank run AMC to NBFC/s?

- **To test whether a bank has direct and indirect exposure to a particular NBFC or multiple NBFCs.**
- **To test whether the indirect exposure is near RBI limits or below RBI limits of 10-15% of NAV.**

5. Financial institutions interdependency

Dwelling further on the joint/dual exposure we consider the interdependency between these financial institutions. Table No.2, displays the net exposure of Banks, AMCs and Insurance companies to NBFCs. One can observe that the net exposure of banks has increased 34% over the period 2012 to 2016. However, a striking observation is the phenomenal (~ 1694%) increase in the net exposure of AMCs to NBFCs during the same period. This increased exposure was attributed to higher NBFC floated commercial paper (CP) and bond issuances during the period 2012 to 2016. Increased demand from AMCs side for the NBFC floated securities (debentures and commercial paper) has gradually shifted their portfolio composition exposure in their liquid and debt funds' portfolio from bank certificate of deposits. Earlier, AMCs portfolio composition for money market and liquid funds included more than 50% bank issued certificate of deposits²⁹. In late 2008, the bank run mutual funds were faced with redemptions to the scale of INR.540 billion in the money market mutual funds. To enable the bank run mutual funds to honor their redemption claim commitments, the RBI provided a repo-facility to the parent banks worth INR.600 billion for a fortnight.³⁰ To curb this interdependency among banks and their mutual fund arm RBI imposed a limit on bank investment in debt oriented mutual funds at 10% of the banks' net worth in May 2011³¹. Owing to this step, bank issued certificate of deposits outstanding have more than halved to INR.1,286.2 billion (October 2017) from INR. 3410.54 billion (Mar 2010³²). Typically, AMCs used to invest 50% - 70% of their portfolio under the liquid and money market fund schemes in bank issued certificate of deposits³³. Gradually, AMCs' shifted their portfolio composition from certificate of deposits to commercial paper in the debt oriented funds such as liquid and money market funds. Commercial paper used to form roughly ~30% of Net asset value (NAV) of a AMCs portfolio composition in 2011 that has increased to 70-80% of NAV in 2017. Similarly, the percentage holdings of bank issued certificate of deposits has decreased from 50-70% to roughly ~10% or less of the Net Asset Value (NAV) of a scheme. Increased investment in commercial paper led to a more than six times increase in commercial paper issuances to 4,892 billion (October 2017) from 760.56 billion (March 2010)³⁴. A major portion of this commercial paper that these AMC floated liquid and money market funds invest into belongs to NBFCs.

A similar trend was also observed in, bond issuances with a major participation from NBFC companies.

²⁹ ACE Mutual Fund database, Analysis of Top 5 money market funds and liquid funds' portfolio composition

³⁰ Mutual Funds and Market Development in India, 07 Jul 2011, available on the internet at, https://www.rbi.org.in/scripts/BS_SpeechesView.aspx?id=584, accessed on 23 Jan 2018.

³¹ RBI Speeches, Mutual Funds and Market Development in India, Jul 2011, available on the internet at, https://www.rbi.org.in/scripts/BS_SpeechesView.aspx?id=584, accessed on 24 Jan 2018

³² RBI Bulletin, Issue of Certificate of Deposits by Scheduled Commercial Banks, Feb 2011, available on the internet at, https://www.rbi.org.in/Scripts/BS_ViewBulletin.aspx?Id=11959, accessed on 23 Jan 2018.

³³ ACE Mutual Fund database, Average portfolio composition of top 5 liquid and MMMF schemes from 2011 to 2017

³⁴ RBI Bulletin, Issue of Commercial Paper by Companies, 2011, available on the internet at, https://www.rbi.org.in/Scripts/BS_ViewBulletin.aspx?Id=11960, accessed on 23 Jan 2018.

Table No. 2: Direct net exposure of Banks, AMCs and Insurance Companies to NBFCs

INR. Billion	March 2012	March 2013	March 2014	September 2014	March 2015	March 2016
Banks (Net)	1513	1453	2919	1495	1595	2029
AMCs (Net)	83	624	756	312	1008	1489
Insurance Companies (Net)	780	880	965	1023	1760	1038

Source: Created by authors, Reserve Bank of India, Financial Stability Report, 2013, 2014, 2015, available on the internet at, <https://rbi.org.in/Scripts/FsReports.aspx>, accessed on 23 Jan 2018

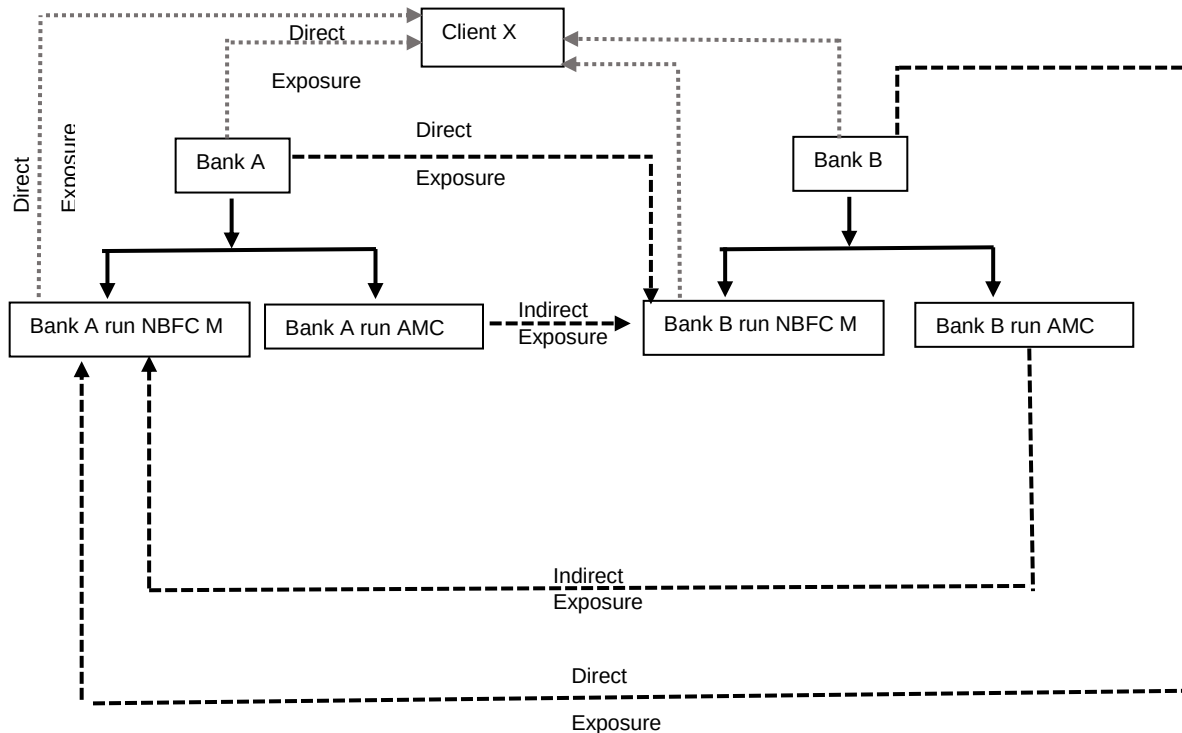
Earlier we discussed about how a parent bank may have exposure to a particular NBFC and its AMC may also have an exposure to that NBFC. This amounts to dual exposure.

Existing banks, corporate houses and financial institutions (other than banks) may set up NBFCs. The NBFCs in our proposal are institutions that lend to retail and corporate clients. Now we discuss how a parent bank and its AMC that have exposure to another bank run NBFC or a corporate run NBFC may create a further web of interdependency in case of bank-run NBFCs. A bank-run AMC may invest in the securities floated by another bank-run NBFC or a corporate run NBFC. For example, Axis liquid fund may have an exposure to Aditya Birla Housing Finance and HDB Financial Services floated commercial paper or debentures. This situation adds to the interdependency among financial institutions such as banks, AMCs and NBFCs among each other. In case of a corporate-run NBFC, the same may have an exposure to multiple banks in terms of bank borrowing and multiple AMC exposure. If a parent bank and its AMC has exposure to such a corporate run NBFC and this NBFC faces issues. This situation will not only impact the parent bank, but also its AMC, the other banks that have lent to the NBFC and other bank run AMCs that have exposure to this particular AMC.

For example, Bank “A” has exposure to corporate run NBFC “I”; Bank “A” also has a AMC “M”. This AMC “M” has also exposure to NBFC “I”. The exposure of Bank “A” and AMC “M” to NBFC “I” account for the joint exposure. Now this NBFC “I” may also have exposure to other bank “B” which may also have its AMC “N”. This makes the joint exposure of bank “B” and AMC “N” to NBFC “I” in the system. In case NBFC “I” witnesses a default in its loans. The Bank “A”, AMC “M”, Bank “B” and AMC “N” may end up having a joint exposure to NBFC “I”. Although the exposure limits of Bank “A” and Bank “B” and AMC “M” and “N” may be within the individual RBI limits. On the whole, it may create a systemic risk for the whole financial system.

This is only one form of interdependency that is explained here. The other form of interdependency may be created when a bank and its AMC have an exposure to another bank run NBFC. This adds manifold to the systemic risk of the financial eco-system. For example, Axis Liquid Fund may have an exposure to ICICI Home Finance. In case ICICI promoted ICICI Home Finance faces NPA, this will not only impact the ICICI Home Finance Promoter, other AMCs that may have an exposure to it, Axis Liquid Fund that may have an exposure to ICICI Home Finance and the parent bank Axis Bank. Hence, the interdependency in case of bank-run NBFCs the systemic risk increases slightly more than a corporate run NBFC.

Diagram No. 3: Financial Interdependency among financial institutions and banks



Source: Created by authors

This leads to a fundamental question, whether AMCs play the role of liquidity transformation in addition to the banks and NBFCs in India. The banks and NBFCs undertake the business of direct lending, whereas AMCs of these banks may be involved in an indirect route of lending through subscription of NBFC commercial papers and debentures. Although Acharya et.al, (2013) has only considered NBFCs as shadow banks and has not addressed asset management companies (AMCs) that might undertake liquidity transformation especially in an open-ended scheme. Here, the AMC may not function as a pass-through but may indulge in the function of liquidity transformation where the scheme may tend to hold cash in excess of their redemption requirements. This may lead to AMCs functioning as shadow banks.

Proposition 2: Do bank run mutual fund arms undertake liquidity transformation through indirect exposure to Non-Banking Financial Companies floated instruments?

- To test whether AMC's undertake liquidity transformation
- A positive relation between a AMC's scheme cash holdings and portfolio composition suggests liquidity transformation. (Sunderam and Cherenko, 2016)
- Liquidity transformation may cause financial stability problems (Goldstein et al (2015), Hanouna et al (2015), SEC (2015), FSOC (2014), Feroliet et al (2014), Chen et al (2010))

This leads to a fundamental question that does the systemic risk increase due to the joint exposure of the bank and its AMC to corporate run and bank run NBFCs. As shown in Diagram No.3 there may be a possibility that the same client may have an exposure to multiple banks and NBFCs and these banks and NBFCs in turn may have an exposure to their AMC's. This will probably lead to a manifold increase the systemic risk faced in the financial system. This interdependency leads us to our final proposition

Proposition 3: Does consolidated bank systemic risk increase using joint exposure to a particular NBFC (bank run or corporate run) through direct bank exposure and indirect bank exposure (through the bank run AMC) to corporate and bank run NBFCs.

- To test whether banks stress testing results will alter significantly when the joint exposure is considered.
- Should regulators such as RBI actively monitor the joint exposure undertaken by banks' and their mutual fund arm to the same corporate and bank run NBFC? At present, the regulators have laid out norms to curb exposure to NBFCs for banks and mutual funds schemes at an individual level. However, they have not considered the joint exposure to the same NBFC through both the bank and its mutual fund arm.

Recently a IMF ((International Monetary Fund) and World Bank FSAP (Financial Sector Assessment Program) report³⁵ and news reports³⁶ have been stressing on how debt mutual fund schemes exposure to NBFCs is posing a systemic risk for the financial system. Most of the scheme exposure lies in the liquid fund category which subscribe to highly rated commercial paper and debentures of these companies. However, the credit ratings are just opinions and may not reflect the true financial picture of the NBFCs in question. The Securities Exchange Board of India (SEBI)³⁷ through its debt funds deployment report in September 2017 indicates that 3.78% of the total debt AUM

³⁵ The Dalal Street Investment Journal, FSAP raises concerns over rising MF exposure to NBFC bonds, 28 Dec 2017, available on the internet at, <http://www.dsij.in/DSIJArticleDetail/ArtMID/10163/ArticleID/32/FSAP-raises-concerns-over-rising-MF-exposure-to-NBFC-bonds>, accessed on 29 Jan 2018.

³⁶ The Economic Times, Debt MFs gorging on NBFC paper pose systemic risk, 27 Dec 2017, available on the internet at, <https://economictimes.indiatimes.com/markets/stocks/news/debt-mfs-gorging-on-nbfc-paper-pose-systemic-risk/articleshow/62262958.cms>, accessed on 29 Jan 2018.

³⁷ SEBI Deployment of Debt Funds Monthly Report September 2017, available on the internet at, <https://www.sebi.gov.in/statistics/mutual-fund/deployment/2017/debtsep17.html>, accessed on 29 Jan 2018.

exposure (INR.9508.3 billion) is to NBFCs commercial paper and 7.19% to NBFC floated bonds and debentures.

6. Data Sources, Methodology and Pilot study

The data for this research proposal is sourced from the Reserve Bank of India database warehouse for NBFCs, Scheduled Commercial banks, RBI Bulletin, Financial Stability reports, Micro Finance Institutions Network (MFIN) publications, NBFC annual and quarterly report, SEBI deployment of debt funds monthly reports, news articles, Thomson Reuters database, Association of Mutual Funds in India (AMFI), Asset Management Companies Monthly Portfolio Disclosure and ACE Mutual Fund Database (Courtesy Indian Institute of Management, Bangalore).

The methodology proposed in this study is the use of General Moment of Methods (GMM) Technique and Least Squares with Bai Perron tests identified L+1 sequentially determined break model. More details about the model are available on the Eviews help guide.³⁸

We have already conducted a pilot study on the liquidity transformation aspect of the research proposition using Bai-Perron tests identified L+1 sequentially determined break model where we find evidence of liquidity transformation being undertaken by bank run AMCs through their liquid schemes. The two liquid schemes considered for this pilot study were ICICI Prudential Liquid Scheme and Axis Liquid Scheme for a period 2009 to 2017. Monthly portfolio data was considered and the relation between cash holdings and the portfolio composition was studied.

The details and results of the pilot study are available in the subsequent section.

³⁸ Eviews help guide, Estimating Least Squares With Breakpoints in eviews, available on the internet at, http://www.eviews.com/help/helpintro.html#page/content/multibreak-Estimating_Least_Squares_with_Breakpoints_in_EVi.html, accessed on 29 Jan 2018.

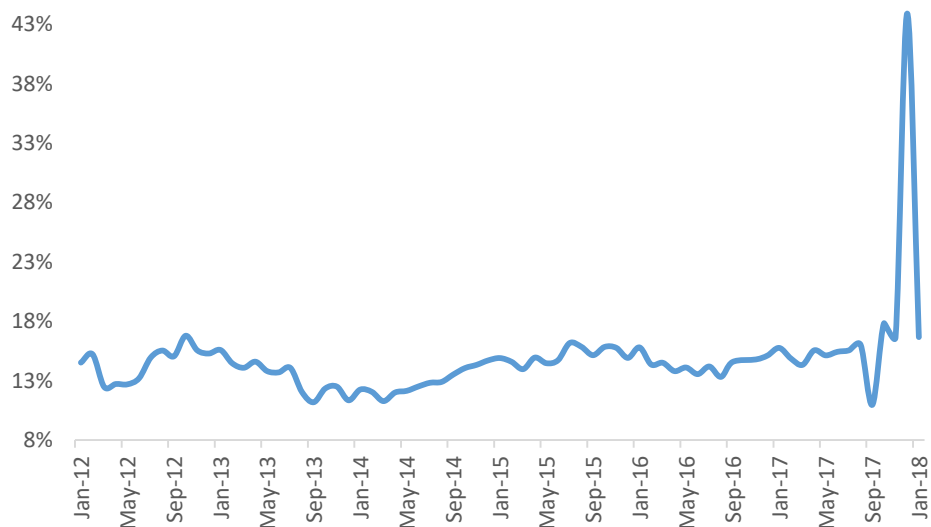
7. Pilot study

In this pilot study we attempt to test whether mutual fund arms undertake liquidity transformation i.e. in debt oriented schemes. These debt schemes include money market mutual funds, liquid funds, GILT funds and debt schemes. Together these form 52% of the total AuM worth INR.21.26 trillion³⁹ on 31 Dec 2017. Liquid and Money market funds form 13%, Gilt forms 1% and debt oriented schemes form 38% of the total AuM. Bank sponsored mutual funds form around 50% or more of the total AuM in the mutual fund industry as on 31 Jan 2018 with ICICI prudential and HDFC mutual fund leading the market with 13% and 12.9% share followed by State Bank of India (SBI) Kotak Mahindra mutual fund and Axis mutual fund at 9.2%, 5.3% and 3.3% respectively.

This pilot study covers: a) overall debt oriented schemes and b) liquid fund of ICICI mutual fund and c) liquid fund of Axis mutual fund. The reason for considering debt schemes is that these debt schemes subscribe to the NBFC floated debentures and commercial paper which forms 44% of the total AuM on 31 Dec 2017 thereby supporting the NBFCs credit requirement.

Diagram No 4: NBFC exposure to Debt schemes Jan 2012 – Jan 2018

This diagram presents the NBFC exposure that debt mutual fund schemes have on an aggregate level from Jan 2012 to Jan 2018. NBFC exposure is calculated as Total amount invested in NBFC floated instruments divided by total AuM.



Source: Securities Exchange Board of India (SEBI), Deployment of Debt funds reports

Under SEBI guidelines the single sector exposure of mutual funds is capped at 30% of the net asset value (NAV) of a scheme i.e. NBFCs (this includes the housing finance companies)⁴⁰. However, in

³⁹ Association of Mutual Funds India (AMFI) AUM – Category /Age wise and folio data, available on the internet at, <https://www.amfiindia.com/research-information/aum-data/age-wise-folio-data>, accessed on 16 Mar 2018.

⁴⁰ IndiaInfoline, The single sector exposure limit stipulated by SEBI for mutual funds is 30% of the net asset value of a scheme for NBFCs, available on the internet at, <https://www.indiaonline.com/article/capital-market-mutual-funds-reports/the-single->

December 2017, this exposure over shot the required statute of 30% by 14% on the whole. This figure was later on brought down to 17% immediately by Jan 2018.

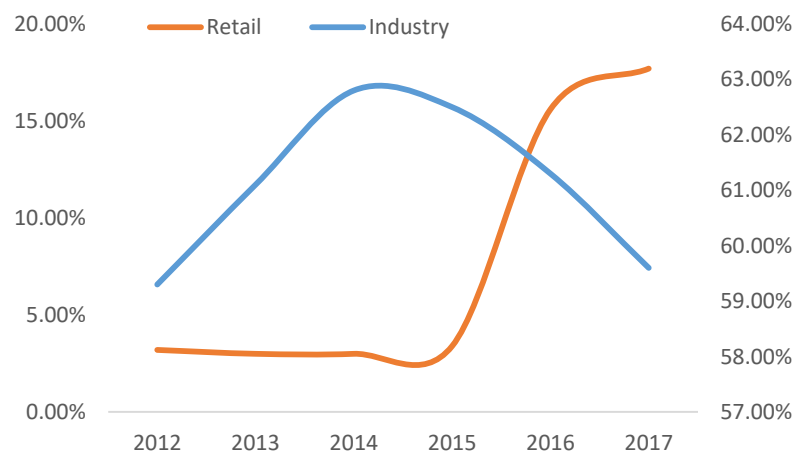
Under the liquid fund and money market mutual fund schemes, the NBFC floated securities form nearly 30% or more of their total AuMs. Although these are investment grade BBB and typically A and above rated. The reason for worry is that even top rated securities default. Ratings are opinions and recent occurrences of default at the hands of Amtek Auto, JSPL and others have left a mark on these mutual funds as they are exposed to these securities. Although, the example belongs to corporate floated securities, one can extend this example to NBFCs as well in two different ways.

- A) Scenario 3 and 4 from the introduction, where a borrower defaults to whom both banks, NBFCs and the mutual fund arm may have an exposure.
- B) The NBFC may default on its securities i.e. commercial paper, debentures and others.

The reason for this stems from the fact that the sectoral deployment of the credit is focused on industry. In 2017, industry formed 59.6% of the total credit deployed by NBFCs.

Diagram No 5: Sectoral deployment of credit by NBFCs

This diagram presents the retail and industry credit deployment as a percentage of the total credit deployment of NBFCs. The left y-axis represents the industry percentages and the right y-axis represents the retail percentages. The x-axis presents the period from 2012 to 2017.



Source: RBI Bulletin October 2017

One observes that the sectoral deployment of the NBFCs like the scheduled commercial banking sector is tilted towards industry. However, since 2015 there is a sudden increase in the lending to the retail segment. With finite industry and retail participants, the possibility of both banks and these NBFCs having an exposure to the same borrower are extremely likely.

For testing whether liquidity transformation is undertaken by AMCs, we first consider the overall debt schemes portfolio composition. Here we look at the cash holding i.e. Collateralized Lending and Borrowing Obligations (CBLO) and T Bills and how they are related to the portfolio of securities held in the mutual fund scheme.

Ha: A positive relation between portfolio securities and cash holding will suggest liquidity transformation undertaken

Ho: A negative relation between portfolio securities and cash holding will suggest no liquidity transformation undertaken

We use a break-point model that employs bai perron L+1 Vs L sequentially determined breaks. The data for the period Jan 2012 to Jan 2018 is considered in this pilot study.

Table 3: Result for breakpoint regression

Dependent variable	CBLO + T Bills		
	Period 1	Period 2	Period 3
Date	Jan 12 – Nov 12	Dec 12 - Mar 17	Apr 17 – Jan 18
C	0.36	0.98***	0.28
Bank Certificate of Deposit	1.19	-0.94***	-0.09
Bank FD	-9.82**	-1.15***	-2.51***
Government Securities	0.70	-0.96***	-0.04
NBFC securities	-1.91***	-1.21***	1.18**
Real Estate	1.86**	-1.23**	9.10
Others	0.53	-0.81***	-0.57*
PSU Bond debt	-1.92***	-1.27***	-2.25**

***, **, * significant at level 1%, 5% and 10% respectively

This table is an indication of the relation between the portfolio and the cash holding of the debt mutual fund schemes. The period is split based on the Bai Perron measure to identify break points. Under this Table 3 data suggests that there exists a positive relation between cash holdings and NBFC securities and Real estate securities held in the portfolio. For every rupee one increase in the nbfc security the scheme has to hold 1.18 rupees extra. On a similar level, the scheme will hold 9.1 rupees for one rupee increase in real estate holding.

However, this doesn't account for the debt fund managers maintaining cash balances based on the previous years trend of redemptions or the past few months redemption patterns or their holding cash till suitable investment opportunities arise or increased returns on the schemes. For the relation between cash holding (i.e. CBLO) and redemptions in the present year and over the past year refer to Table 1 in appendix.

All these factors have to be controlled for an unbiased result of the relation between cash holdings and portfolio securities.

Table 4: Result for least square regression with control for redemption in past year

Dependent variable	CBLO + TBills
Date	Jan 2012 – Jan 2018
C	-0.001
Bank Certificate of Deposit	0.03
Bank FD	-0.15
Government Securities	0.37**
NBFC securities	0.29
Real Estate	-0.38
Others	-0.13
PSU Bond debt	0.79**
R-square	70.6%
Adjusted R	67.3%

Table 5: Result for breakpoint regression with control for redemption in past year

Dependent variable	CBLO + T Bills (t)			
	Period 1	Period 2	Period 3	Period 4
	Jan 12 – Jul 13	Aug 13 – Feb 15	Mar 15 – Jun 16	Jul 16 – Jan 18
C	0.00	0.00	0.00	0.00
Bank Certificate of Deposit	0.03	-0.36**	-0.48***	0.56
Bank FD	-2.94***	0.35	0.63	-3.50
Government Securities	-1.24**	-0.34	0.02	1.06***
NBFC securities	1.30**	3.2**	0.22	0.04
Real Estate	59.96***	-13.16	9.67***	-0.63
Others	0.001	-0.96***	-0.20	0.11
PSU Bond debt	-0.07	1.97	1.09**	-0.47***
R-square				90%
Adjusted R				83.6%

Under this Table 5 data suggests that there exists a positive relation between cash holdings and NBFC securities across the periods. However, Real estate securities held in the portfolio show a positive and negative relation with cash holdings intermittently. For every rupee one increase in the nbfc security the scheme has to hold 1.30, 3.2, 0.22 or 0.04 rupees extra. On a similar level, the scheme will hold 59.6 and 9.67 rupees for one rupee increase in real estate holding. Another aspect is the positive relation displayed with PSU bonds, bank FD and government securities. This may indicate that these are not as liquid as they may appear. However, this needs to be further evaluated based on the credit spreads of PSU bonds and government securities during the periods when the relation between these turns positive. This may suggest that there may exist pressure on the liquidity of these securities.

Conclusion:

We began with developing a framework to study how financial interdependency among banks and NBFCs may increase the risk exposure of banks. We developed four scenarios that gradually led to us to build that a bank may face increased risk as it may face exposure to NBFCs through:

- a) Direct lending to the NBFC
- b) The bank sponsored mutual fund investing in debt securities and commercial paper issued by the NBFC
- c) Exposure to common clients

This implies developing a model where one can estimate the overlaps of a NBFC that exist between a bank sponsored mutual fund and the bank itself. A drawback is that NBFCs and banks don't disclose the common clients hence it is difficult to estimate the exact exposure. However, it is necessary to consider this common client exposure and the financial interdependency to arrive at the total exposure of the banks.

Another aspect of this study is to explore whether mutual fund arms undertake liquidity transformation through their debt oriented schemes. In case a mutual fund functions as a credit intermediation agency then it will hold more cash with respect to its average. On an average the debt oriented schemes held cash more than the average of 4.41% - 28 times out of 72 observations i.e. 39% of the total observations. A few possible reasons are the large inflows of cash into the system and the fund managers' penchant to hold cash till he/she finds a suitable investment or expectation of cash outflows - redemptions.

To study whether AMCs undertake liquidity transformation, we develop a simple model (Cherenko and Sunderam, 2016) where if a positive relation exists between cash holdings and the portfolio this is indicative of liquidity transformation. We conducted a pilot study using the data of debt deployment from Jan 2012 to Jan 2018 and applied a break point regression through which the results in table 4 and 5 suggested that there exist a few securities in the portfolio that display a positive relation with the cash holdings.

Upon controlling for redemptions in the study and again running the model we still find that during certain periods this positive relation between NBFC floated securities, Other Bank floated securities and government and PSU floated securities and cash holdings exists. This suggests some form of liquidity transformation at the base level. However, this cash holding and a few other control variables are required to be certain of the results.

In case there exists credit intermediation function especially for NBFC and Real estate exposure the banks may have an increased risk. NBFCs already undertake the credit intermediation in the financial system, hence routing short-term loans through a bank sponsored mutual fund arm may well bypass the regulatory requirements to some extent, however, it may expose a bank to a great risk especially if both the bank and the NBFC share the finite industry borrowers set.

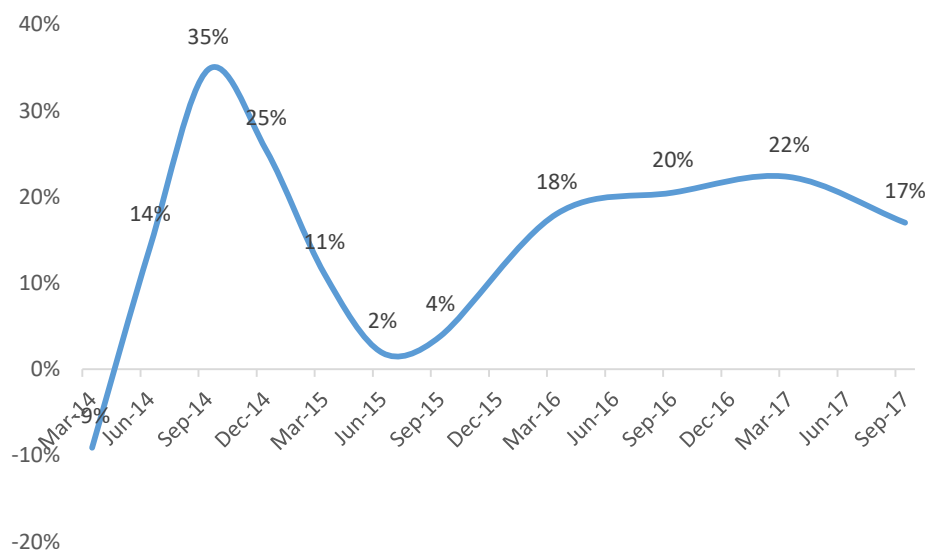
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Appendix

Appendix Diagram 1: Rolling correlation between NBFC loan growth and Scheduled Commercial Bank loan growth

The rolling correlation between NBFC loan growth and scheduled commercial bank loan growth is presented for the quarter ended March 2013 to September 2017. This rolling correlation is calculated for the period beginning March 2013 to March 2014. Subsequently, the next quarter is added to the correlation calculation. June 2014 correlation estimate considers March 2013- June 2014 growth figures.



Source: Created by authors

Diagram 1 presents the rolling correlation between NBFC quarterly loan growth (y-o-y) and Bank credit quarterly growth (y-o-y). The correlation is weak positive across the period March 2014 to Sep 2017. This indicates that the existing banking sector and the NBFCs serve as complements to each other. However, there are periods when they tend to serve as substitutes for lending but these periods are short lived.

Appendix Table 1

Table 1 presents the results of a least squares regression with lagged regressors of the variable redemption (t) as a percentage of total AuM (t).

Dependent variable	CBLO (t)		
C	0.04***	C	0.04***
Redemption (t-1)	-0.04	$\sum$ Redemption (t-12) from (t-24)	0.09*
Redemption (t-2)	0.05	$\sum$ Redemption (t-13) from (t-25)	0.02
Redemption (t-3)	0.11**	$\sum$ Redemption (t-14) from (t-26)	-0.06

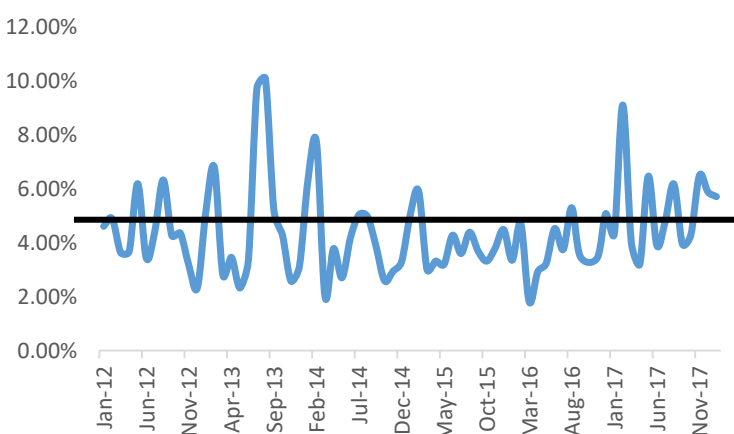
Redemption (t-4)	0.08*	$\sum$ Redemption (t-15) from (t-27)	0.03
Redemption (t-5)	-0.04	$\sum$ Redemption (t-16) from (t-28)	-0.04
Redemption (t-6)	-0.03	$\sum$ Redemption (t-17) from (t-29)	-0.06
		$\sum$ Redemption (t-18) from (t-30)	-0.01
R square	19.3%		21.8%
Adjusted R square	11.1%		12.4%

Source: Created by authors

The table indicates that the CBLO (i.e. representative cash holding) has a positive significant relation with redemptions (t-3) and (t-4). Hence, this suggests that a fund manager tends to hold more cash when there are higher cash flows. This is consistent with the understanding that when cash inflows increase in the debt oriented schemes the cash holdings increase. The level of cash held is dependent on the previous cash flows. On the right side of the table in column 3 and 4, we find the relation between the redemptions for the year i.e. from (t-24) to (t-12) also display a positive significant relation with CBLO.

Continuation to this is the diagram 2 below which shows that the average CBLO held in these debt oriented schemes was 4.41% from the period Jan 2012 to Jan 2018. Of the 72 observations, 28 times the cash holding was above the average held $\sim 4.41\%$. The cash holdings was increased when there was a redemption outflow was higher either during the month (t) or (t-1) or (t-2). Hence, this suggests that a fund manager undertook to maintain more cash balances when there were large outflows in the previous months.

Appendix Diagram 2: CBLO in Debt oriented schemes Jan 2012 to Jan 2018



Source: Created by authors